



Genome-wide identification of trihelix transcription factors in the apple genome in silico

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Abstract

Trihelix transcription factors are involved in the growth and development of plants, as well as various stress responses. In the study presented, we identified 37 trihelix family genes in the apple genome (MdTH). The trihelix genes were located on 13 chromosomes. Phylogenetic analysis of these MdTH and the trihelix genes of other species divided them into six sub-families: GT-1, GT-2, SH4, SIP1, GT γ , and GT δ . The genes of different groups significantly diverged in their gene structure and conserved functional domains. Cis-element analysis showed that promoter sequences of MdTH genes contained light response elements, phytohormone response elements, and stress-related cis-elements. The expression pattern analysis results demonstrated that MdTH were regulated by drought, salinity, as well as high and low temperatures. MdTH4 and MdTH24 were highly regulated by soil salinity, MdTH4—by drought. MdTH30 showed high expression under low temperature; MdTH8, MdTH20, and MdTH36—under high temperature.

Keywords *Malus* · Trihelix transcription factors · Expression analysis · Stress

Introduction

Trihelix transcription factors were identified only in plants (Mukherjee et al. 2019). They were first identified and isolated in pea in 1990s. They bind to the core sequence 5'-G-Pu-(T/A)-A-A-(T/A)-3' of the promoter region of the *rbcS-3A* gene involved in the regulation of light-dependent expression (Green et al. 1987). This family was originally named GT for the ability of transcription factors to bind to light-responsive GT elements. The DNA-binding domain of these transcription factors has typical three-helix structure (helix-loop-helix-loop-helix), which lies behind the name “trihelix transcription factor.” Studies have shown that the trihelix structure of GT factors is highly similar to the structure of Myb/SANT-LIKE-DNA-binding domains (Nagano 2000). In the plant genome, the GT factors evolved from Myb/SANT-LIKE proteins. The gaps between the helices in trihelix transcription factors form a DNA-binding structure,

which is different from Myb/SANT-LIKE proteins (Nagano 2000; Qin et al. 2014). According to the data provided in the protein domain databases such as Pfam, the Myb/SANT-LIKE domain is a conserved domain of trihelix transcription factors.

The trihelix family genes have been identified in many plants such as *Arabidopsis*, tomato, chrysanthemum, rice, and other (Kaplan-Levy et al. 2012; Yu et al. 2015; Song et al. 2016; Li et al. 2019). Thus, in *Arabidopsis*, 30 family members of several subfamilies (GT-1, GT-2, GT γ , SH4, and SIP1) were identified (Kaplan-Levy et al. 2012; Gao et al. 2009). Thirty-six trihelix family genes were identified in tomato. They were classified into six subfamilies (GT-1, GT-2, SH4, SIP1, GT- γ , and GT- δ) (Yu et al. 2015). The last family was not found in *Arabidopsis*. In the rice genome, 41 trihelix family genes, which are the members of five sub-families (SIP1, GT γ , GT, SH4, and GT δ), were identified (Li et al. 2019).

Some studies have reported the involvement of trihelix genes in complex physiological functions. In rice, trihelix genes may be involved in abiotic stress (drought, salinity) and up-regulated by signaling molecules, such as abscisic acid and hydrogen peroxide (Li et al. 2019). In *Arabidopsis*, the GT-1 subfamily genes may be involved in salt stress and pathogen infection response (Murata et al. 2002). In

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soybean seedlings, GmGT-2A and GmGT-2B transcription factors are induced by abscisic acid, drought, high salt levels, and cold (Xie et al. 2009). The *Arabidopsis thaliana* GTL1 (GT2-LIKE1) loss-of-function mutation resulted in a fewer stoma compared to wild-type plants, and therefore, they were more resistant to drought due to less water loss (Yoo et al. 2010). In rice, the *OsGTγ-1* expression increased by 2.5–10 times in response to salt stress; it could also be induced by abscisic acid (Fang et al. 2010). However, in chrysanthemum the expression of some trihelix genes decreased as a result of exposure to abscisic acid (Song et al. 2016). Trihelix genes are also associated with plant morphogenesis. In *Arabidopsis*, the trihelix transcription factor PETAL LOSS (PTL) determines the number of petals per flower and sepal fusion (Brewer et al. 2004). The rice SH4 subfamily gene promotes the development and functioning of the abscission layer in mature seed peduncles (Li et al. 2006). Trihelix transcription factors play an important role in plant responses to abiotic and biotic stresses and developmental processes (Kaplan-Levy et al. 2012).

Apple is one of the most widely cultivated of fruit trees. Its yield depends on many factors, including temperature, light, amount of water, and soil composition. The identification and study of genes involved in developmental processes and a stress response may contribute to a better understanding of molecular mechanisms providing plant growth and development.

In this study, trihelix family genes were identified in silico in the apple variety Golden Delicious. The chromosomal distribution, gene structures, hypothetical protein characteristics, conserved motif composition, cis-elements, phosphorylation sites, and miRNA targeting of the identified trihelix genes were analyzed, including their response to abiotic stress (drought, high and low temperatures). Our results provide a theoretical basis for the future functional characterization of trihelix genes in apple, especially in abiotic stress responses.

Materials and methods

Identification and analysis of trihelix family members in the apple tree

The nucleotide and hypothetical protein sequences of apple genes (*Malus x domestica*, Golden Delicious variety) were retrieved from the Genome Database for Rosaceae (<https://www.rosaceae.org/>); chromosome sequences—from the NCBI GenBank. The protein HMM model of the Myb/SANT-LIKE (PF13837) domain was downloaded from the PFAM database (El-Gebali et al. 2018). It was used to search for matches among the

apple hypothetical protein sequences using HMMER3 (<http://hmmer.org/>). The identified potential matches were tested by scanning with SMART (<http://smart.embl-heidelberg.de/>) (Letunic et al. 2020). The sequences, containing the Myb/SANT-LIKE domain, were considered as trihelix proteins. The alignment of trihelix sequences was performed using the BLAST algorithm (Altschul et al. 1997). The online service Circoletto (Darzentas 2010), which is based on Circos (Krzywinski et al. 2009), was used to visualize the sequence similarity. The exon-intron structure of genes and binding sites of the transcription factors were generated using the Gene Structure Display Server v.2.0 (Hu et al. 2015).

Analysis of hypothetical MdTH proteins

The molecular weights and isoelectric points of MdTH were estimated using the ExPASy Bioinformatics Resource Portal: <https://www.expasy.org/> (Gasteiger et al. 2003). Subcellular localization was predicted using the Localizer 1.0.4 (<https://localizer.csiro.au/cgi-bin/script.py>) (Sperschneider et al. 2017).

Cis-elements analysis of MdTH genes

The 2000-bp upstream sequences of MdTH genes were used for the identification of *cis*-regulatory elements with the PlantCARE database (Lescot et al. 2002). Visualization of search results for the transcription factor binding sites was generated using the Gene Structure Display Server v.2.0 (Hu et al. 2015).

Prediction of miRNA targets and MAPK-specific phosphorylation sites

Potential miRNA targets were calculated using the psRNA-Target online server (<https://www.zhaolab.org/psRNATarget/>) (Dai et al. 2018) with the following parameters: the maximum likelihood—2, UPE (maximum energy to unpair the target site)—less than 25. At the time of analysis, the database contained 207 apple miRNAs.

MAPK-specific phosphorylation sites were predicted by the MusiteDeep tool (<http://musite.net>) (Wang et al. 2020) with a default specificity level.

Multiple sequence alignment and phylogenetic analysis

The amino acid sequences of *Arabidopsis* trihelix transcription factors were loaded from the GenBank database according to the list presented in Kaplan-Levy et al. (2012). The sequences were aligned using the ClustalW algorithm. The maximum

likelihood phylogenetic tree with 1000 bootstrap replicates was constructed using MEGA11 (Tamura et al. 2021).

Plant materials and qRT-PCR

The apple clonal rootstocks MM-106 were used in this study. They have medium frost resistance and low drought tolerance. The rootstocks were grown in the greenhouse with a 16/8-h light/dark photoperiod at 22 °C. The plants were divided into five groups, and one of the groups, the control, was under the conditions described above. The second group was exposed to drought by removing plants from the ground and then placing them on the filter paper; the third group was exposed to high temperature (40 °C) and the fourth to low temperature (4 °C). The plants of the fifth group were watered with a salt solution at a concentration of 0.2 mol/L. Each group was represented by three trees. The light regime in each group was identical.

Leaf samples were collected at the 0, 2nd, 4th, and 24th hour of exposure. Collection at the 0 h was carried out immediately after subjecting plants to stress factors. The collected leaves were immediately frozen in liquid nitrogen.

RNA extraction from frozen leaves was performed using the CTAB method (Jaakola et al. 2001). The quality of extracted RNA was assessed using agarose gel electrophoresis. The concentration of the RNA obtained was measured using the NanoDrop device (ND-8000 Spectrophotometer, Thermo Scientific). RNA was purified from DNA using the DNase I (RNase-free) reagent (Thermo Scientific, USA), according to the protocol. Construction of the cDNA minus strand was performed using the RevertAid H Minus First Strand cDNA Synthesis Kit (Thermo Scientific, USA), according to the manufacturer's protocol.

Primer sequences for qPCR are presented in Supplementary Table 1. Primer efficiency was assessed by the qPCR reaction on the cDNA matrix of the MM-106 apple variety, and the results obtained demonstrated that the efficiency of each pair of primers is within the range of acceptable values of 90–110% (Bustin et al. 2009). The *Ef-1a* gene was chosen as the internal control (Zhao et al. 2012). As the negative control, instead of cDNA, the equal amount of deionized water was used. To analyze the expression of MdTH genes in response to abiotic stress factors, three biological repeats were used in each experimental condition. Thermal cycling and detection were performed in the CFX96 real-time PCR detection system (Bio-Rad, USA). Thermal cycling conditions were set as follows: 95 °C for 5 min followed by 38 cycles of 95 °C for 20 s, 58 °C for 20 s, and 72 °C for 20 s. The calculation of a relative expression level of the genes under

study was carried out using the $2^{-\Delta\Delta CT}$ method (Rao et al. 2013).

Results

Identification of trihelix genes in apple

We used HMMER3 software and the HMM profile from the PFAM database to search for trihelix domains in the apple genome. We then confirmed the presence of these domains using the SMART database, which verified that putative genes contained the Myb/SANT-LIKE domain.

In total, 37 trihelix genes (marked as MdTH) were identified in the genome of the apple variety Golden Delicious. The trihelix proteins in apple were found to have variations in length ranging from 278 to 917 amino acid residues (514.5 on average). The molecular weight of MdTH proteins ranges from 31.9 to 101.3 kDa (57.7 kDa on average). The estimated isoelectric points range from 5.23 to 9.78 (7.4 on average). The results of in silico cellular localization prediction of MdTH proteins demonstrate that 28 out of them are located in the nucleus and 2 in chloroplast (Table 1).

The genome of most domestic apple varieties, including Golden Delicious, contains 17 pairs of chromosomes with the total length of 742.3 Mb (Velasco et al. 2010). Evaluation of the chromosomal distribution of MdTH showed that 13 out of 17 chromosomes (except for 1, 8, 9, and 15) contained at least one of such genes. The largest number of genes (six) is located on 5th and 6th chromosomes, while chromosomes 3, 7, 13, 16, and 17 contained only one gene each. There is no clear relationship between the chromosome length and the number of genes located on it (Fig. 1). Genes located on different chromosomes may have more similarities than those on the same chromosome evidencing that polyploidization processes played a role in the formation of the apple genome. The domesticated apple genome was formed as a result of the duplication and subsequent reorganization of 9 chromosomes of a precursor species (Velasco et al. 2010). A model for the origin of the *M. × domestica* genome designed on the basis of sequencing results suggests that chromosomes 5 and 10 originated from chromosome I of an ancestral species. This explains a high degree of similarity in the structural organization of MdTH8 and MdTH21, MdTH9 and MdTH22, MdTH10, and MdTH23 genes (Fig. 2). Chromosomes 2 and 7, 4 and 12, 6 and 14, 12 and 14, or their individual regions are also of common origin (Velasco et al. 2010). As a result, we can observe that the MdTH genes located on chromosomes 2 and 7, 4 and 12, 6 and 14, 12 and 14, as well as on chromosomes 5 and 10, are more similar to each other than other genes of this family (Fig. 2).

Table 1 Trihelix family members identified in the apple tree

Gene name	Gene ID	Protein length (a.a.)	MW (kDa)	PI	Localization of cells	Chromosome of the gene localization
MdTH1	MD02G1219900	366	40481.75	9.16	Nucleus	2
MdTH2	MD02G1244500	491	53176.73	8.57		2
MdTH3	MD02G1247700	569	65668.97	6.06	Nucleus	2
MdTH4	MD02G1318400	742	81180.12	9.75	Nucleus	2
MdTH5	MD03G1089900	674	74397.21	9.06	Nucleus	3
MdTH6	MD04G1154600	287	34785.25	7.82		4
MdTH7	MD04G1221900	658	74968.15	5.84	Nucleus	4
MdTH8	MD05G1024300	448	51480.28	6.12	Nucleus	5
MdTH9	MD05G1024400	665	74379.05	5.66	Nucleus	5
MdTH10	MD05G1024500	572	64620.71	8.07	Nucleus	5
MdTH11	MD05G1174500	291	32988.47	9.33	Nucleus	5
MdTH12	MD05G1322900	474	53022.84	9.78	Nucleus	5
MdTH13	MD05G1361500	412	46153.60	5.61	Nucleus	5
MdTH14	MD06G1021000	288	33181.41	8.88		6
MdTH15	MD06G1046900	353	39091.11	9.32	Nucleus	6
MdTH16	MD06G1127800	764	83305.38	5.75	Nucleus	6
MdTH17	MD06G1143600	917	100966.22	7.62	Chloroplast	6
MdTH18	MD06G1172900	485	55103.10	6.50		6
MdTH19	MD06G1196500	338	37845.55	7.74	Nucleus	6
MdTH20	MD07G1068800	566	64781.83	5.69	Nucleus	7
MdTH21	MD10G1024800	447	51549.42	6.11	Nucleus	10
MdTH22	MD10G1025100	664	74452.43	6.47	Nucleus	10
MdTH23	MD10G1025200	563	63544.23	8.56	Nucleus	10
MdTH24	MD10G1163600	278	31889.18	7.72	Nucleus	10
MdTH25	MD10G1338800	130	15328.55	9.79	Nucleus	10
MdTH26	MD11G1017600	593	65658.60	9.21	Nucleus	11
MdTH27	MD11G1079000	497	54971.24	6.35	Nucleus	11
MdTH28	MD12G1018900	538	60775.83	6.36	Nucleus	12
MdTH29	MD12G1168300	296	35422.79	8.35		12
MdTH30	MD12G1238000	656	73982.06	5.70	Nucleus	12
MdTH31	MD13G1109800	365	42097.23	6.21	Nucleus	13
MdTH32	MD14G1016900	532	60326.45	6.48	Nucleus	14
MdTH33	MD14G1058900	674	74986.94	9.10		14
MdTH34	MD14G1143600	727	79127.51	5.23	Nucleus	14
MdTH35	MD14G1158900	917	101294.49	8.36	Chloroplast	14
MdTH36	MD16G1109700	382	43571.81	6.05		16
MdTH37	MD17G1017400	417	46616.93	5.47	Nucleus	17

Phylogenetic analysis of trihelix proteins

In order to analyze phylogenetic relationships of trihelix family genes in the apple and the model plant *Arabidopsis thaliana*, an unrooted phylogenetic tree was constructed using 33 trihelix hypothetical proteins of *Arabidopsis thaliana* and 37 of the apple (Fig. 3). MdTH are grouped into six subfamilies (GT-1, GT-2, SH4, GT γ , GT δ , and SIP1) depending on the conserved amino acids of the GT domain, the number of DNA-binding motifs, and the classification of homologs.

GT-2 is the largest subfamily. This cluster includes 14 apple and 6 *Arabidopsis* genes. The SH4 cluster is represented by 13 genes and 8 out of them are apple genes. Fifteen genes are included in the SIP1 cluster and five of them are apple genes. The GT-1 cluster combines ten genes and six out of them are apple genes. The GT γ cluster is represented by five genes, including two MdTH. The GT δ cluster is represented by two apple genes. The genes of this subfamily were not originally identified in the *Arabidopsis* genome (Gao et al. 2009). They were found in the genomes of tomato, sunflower, and rice (Song et al. 2021; Yu et al. 2015; Li et al. 2019).

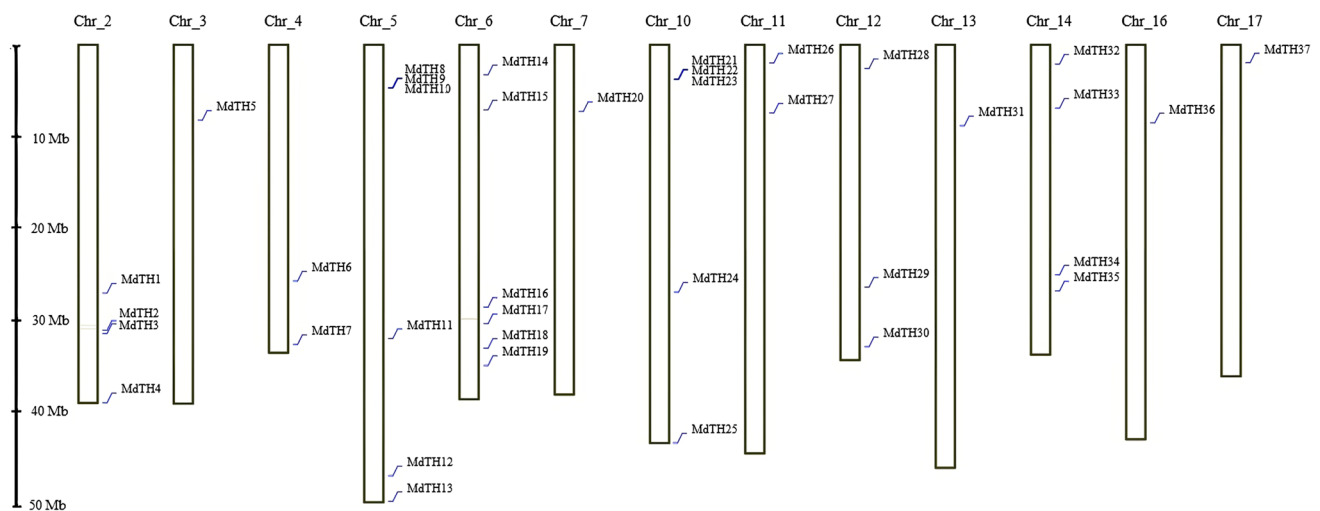
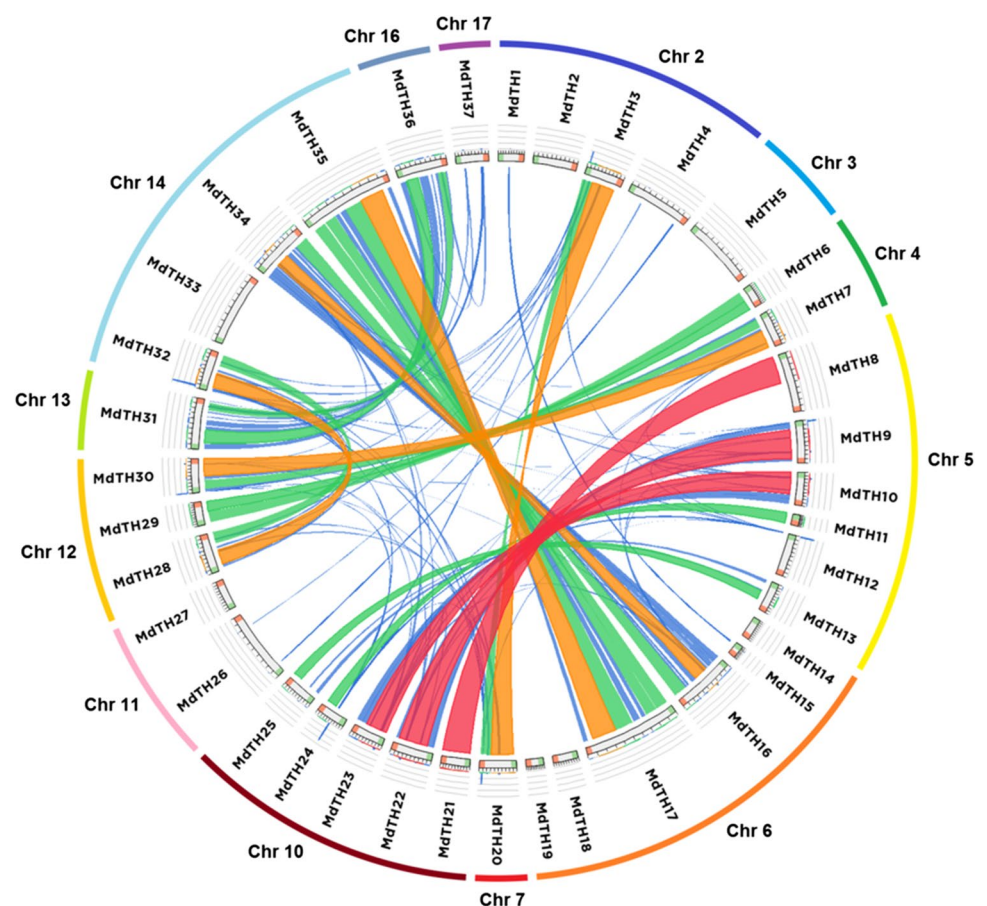


Fig. 1 Distribution of trihelix genes in apple tree chromosomes

Fig. 2 The schematic representation of segmental duplications of apple trihelix genes. The chromosome number is given at the top of the circle segment. The segment's length is proportional to the number and length of the trihelix genes present in each chromosome. The lines inside the circle connect the homologous regions of genes. The thickness of lines is determined by the length of the homologous regions of genes. The color of lines reflects the degree of gene sequence homology, from minimum to maximum (colored according to the score: blue → green → orange → red)

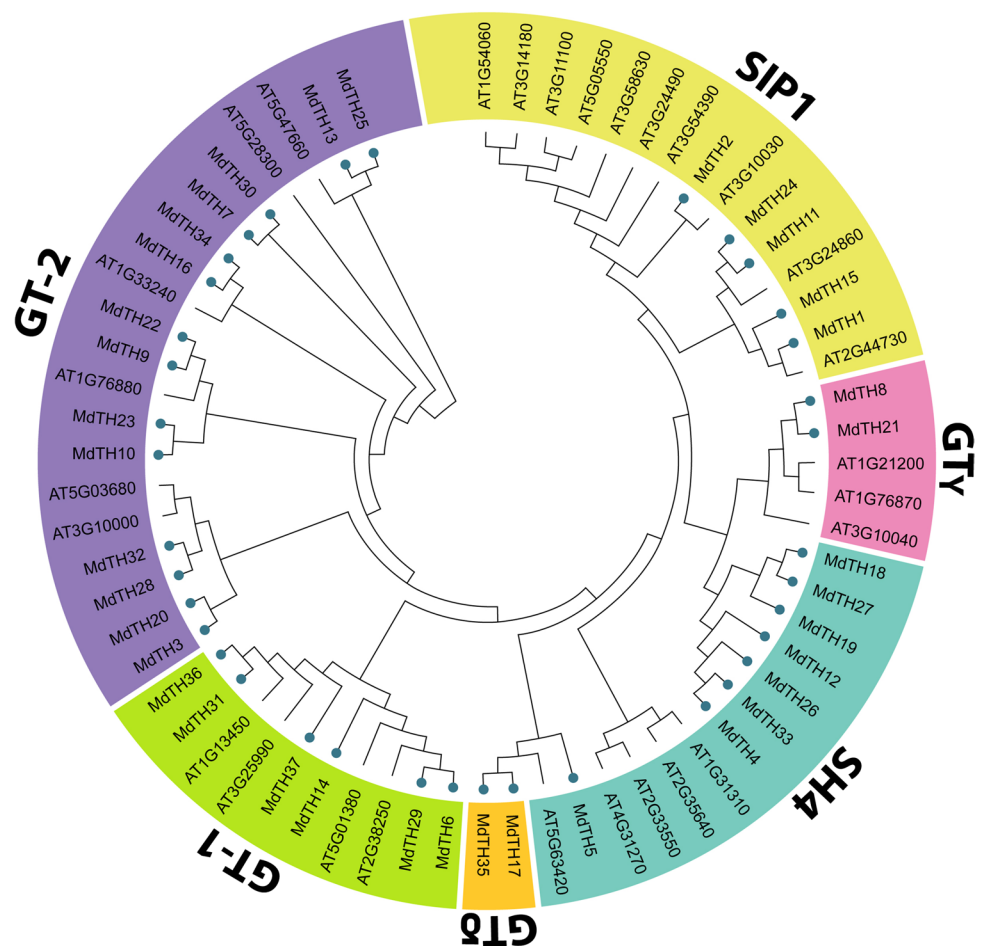


Analysis of the MdTH gene structure and motif

The MEME search tool was used to predict 10 conserved amino acid motifs of MdTHs to analyze their diversity (Fig. 4A). Sequences, containing similar motifs, are grouped

into clusters; the location of motifs correlates with phylogenetic analysis results (Fig. 3, Fig. 4A). Analysis of conserved motifs within the subfamilies showed that motifs 2, 3, 6, and 8 are found only in GT-2 subfamily members. Representatives of the GT-2 subfamily contain the largest number of

Fig. 3 The phylogenetic tree based on the amino acid sequence of apple (Md) and *Arabidopsis thaliana* (AT) trihelix proteins. The maximum likelihood tree was constructed using MEGA11 (bootstrap value = 1000). The clade in green represents GT-1 family, violet represents GT-2 family, yellow represents SIP1 family, pink represents GT γ family, blue represents SH4 family, and orange represents GT δ family



motifs. Four of them have all ten motifs, two have nine, and another two contain eight. GT-2 and GT-1 subfamily members contain motifs 5 and 10.

The most common is motif 1 occurring in all sequences. Motif 7 was found in 36 out of 37 sequences, each containing one copy of it. Motif 4 is the next most common motif and it was found in 32 proteins. Motif 6 was identified in 6 sequences, which are GT2 subfamily members. Six sequences, which are the members of two subclasses, GT-2 and GT-1, contain motif 10. The SH4 subfamily was characterized by the smallest number of conserved motifs—from 1 to 3.

The coding sequences of most MdTH genes are typically interrupted by introns, with the number of introns varying from 1 to 16 among these genes (Fig. 4c). The number of introns in GT-1, GT-2, and SH4 subfamily members ranges from 1 to 8. The intron length and the location in the gene are different in the representatives of different subfamilies and within the subfamilies themselves. The greatest differences from other subfamilies in the exon-intron organization were found in MdTH17 and MdTH35 genes of the GT δ subfamily. Both representatives of the GT δ subfamily have 16 introns. Seven genes are intronless—MdTH8 and

MdTH21 from GT γ subfamily. The MdTH25 of the GT-2 subfamily is also intronless. Four out of five members of the SIP1 subfamily lack introns, while the remaining MdTH2 member has six.

Identification of hypothetical cis-elements in MdTH promoter regions

Analysis of the nucleotide structure of the regions preceding the MdTH genes showed that they are involved in various signaling networks, which ensure the growth and development of plants. One of the key factors influencing the expression of MdTH genes is light. The elements involved in light-dependent signaling pathways were found in the promoter regions of all family members with no exceptions (Fig. 5). Among them, G-box (detected in the promoter regions of 33 out of 37 genes, which is 89.2%), Box 4 (78.4%), GT-1 (64.9%), TCT-motif (56.8%), and AE-box (51.3%) are the most frequently occurring. The elements of the 3-AF1 binding site, AAAC-motif, A-box, ACA-motif, ACE, AT1-motif, ATC-motif, ATCT-motif, Box II, CAAT-box, CAG-motif, chs-CMA1a/2a, chs-Unit 1 m1, GA-motif, Gap-box, GATA-motif, G-Box, GT1-motif, GTGCG-motif,

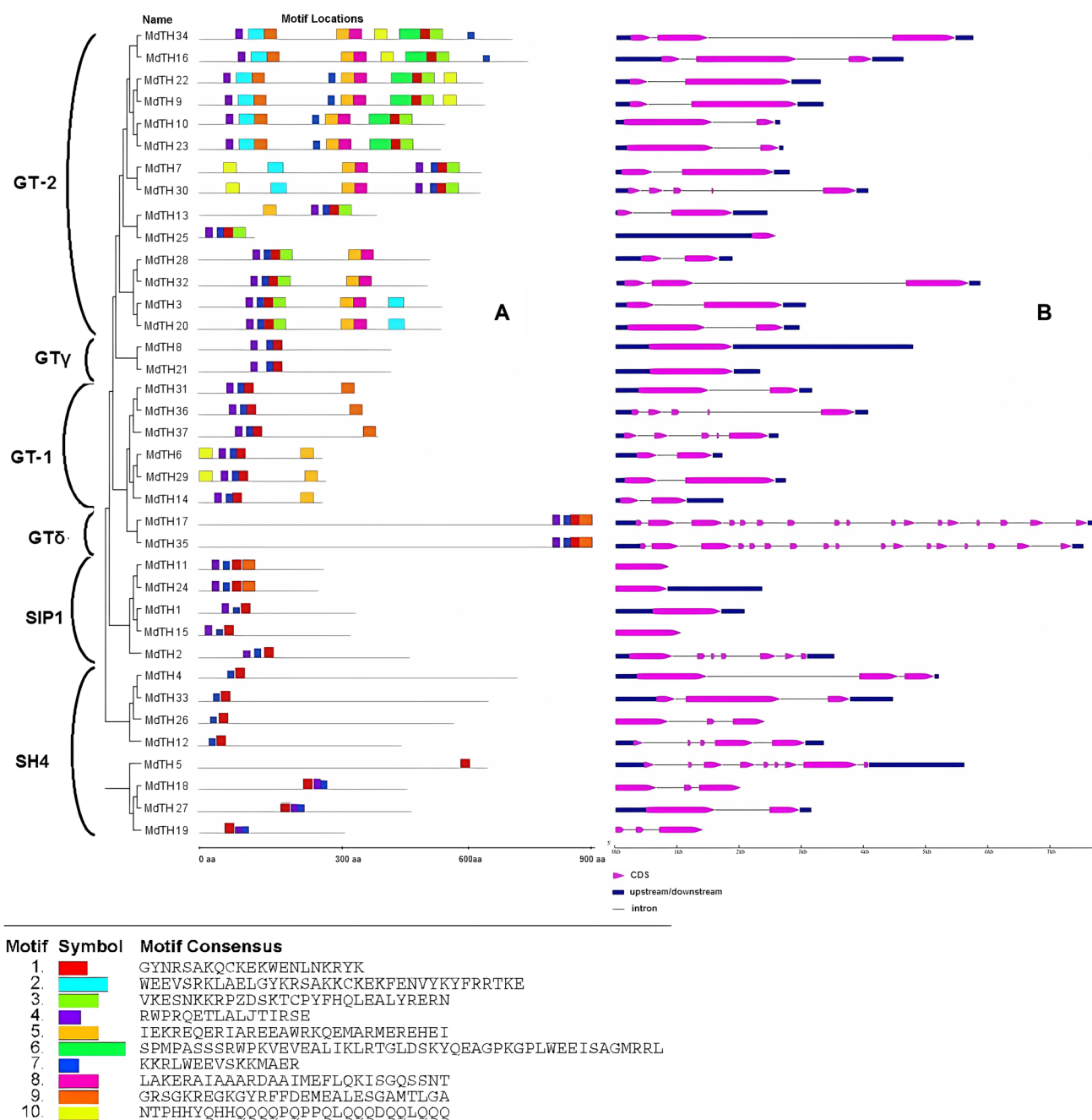


Fig. 4 Apple trihelix gene structure and motifs. **A** The phylogenetic tree based on the amino acid sequence of 37 apple trihelix proteins. The maximum likelihood tree was constructed using MEGA11 (the bootstrap value = 1000). The MEME Suite was used to predict

motifs. The motifs are represented by the boxes of different color. **B** Gene structures. Intron, exon and upstream/downstream sequences are represented by black lines, red boxes and blue boxes respectively.

I-box, LAMP-element, L-box, MRE, Sp1, TCC- motif, and TCCC-motif were also found.

Expression of MdTH genes under stress is evidenced by detected STRE elements (91.9% of the promoter regions contain them) and TC-reach repeats (29.8%). ARE elements involved in the response to anaerobic stress were found in the regulatory regions of 81.1% genes; CG

involved in the response to anoxia in 21.6%; MBS, DRE1, and DRE, regulating the expression of genes under the drought conditions, in 51.4%, 5.4%, and 13.5%, respectively; LTR was involved in the response to low temperatures in 48.6%. The possible expression of MdTH genes in response to wound stress and pathogen attack can be inferred from the presence of WUN, Box S, WRE3 and

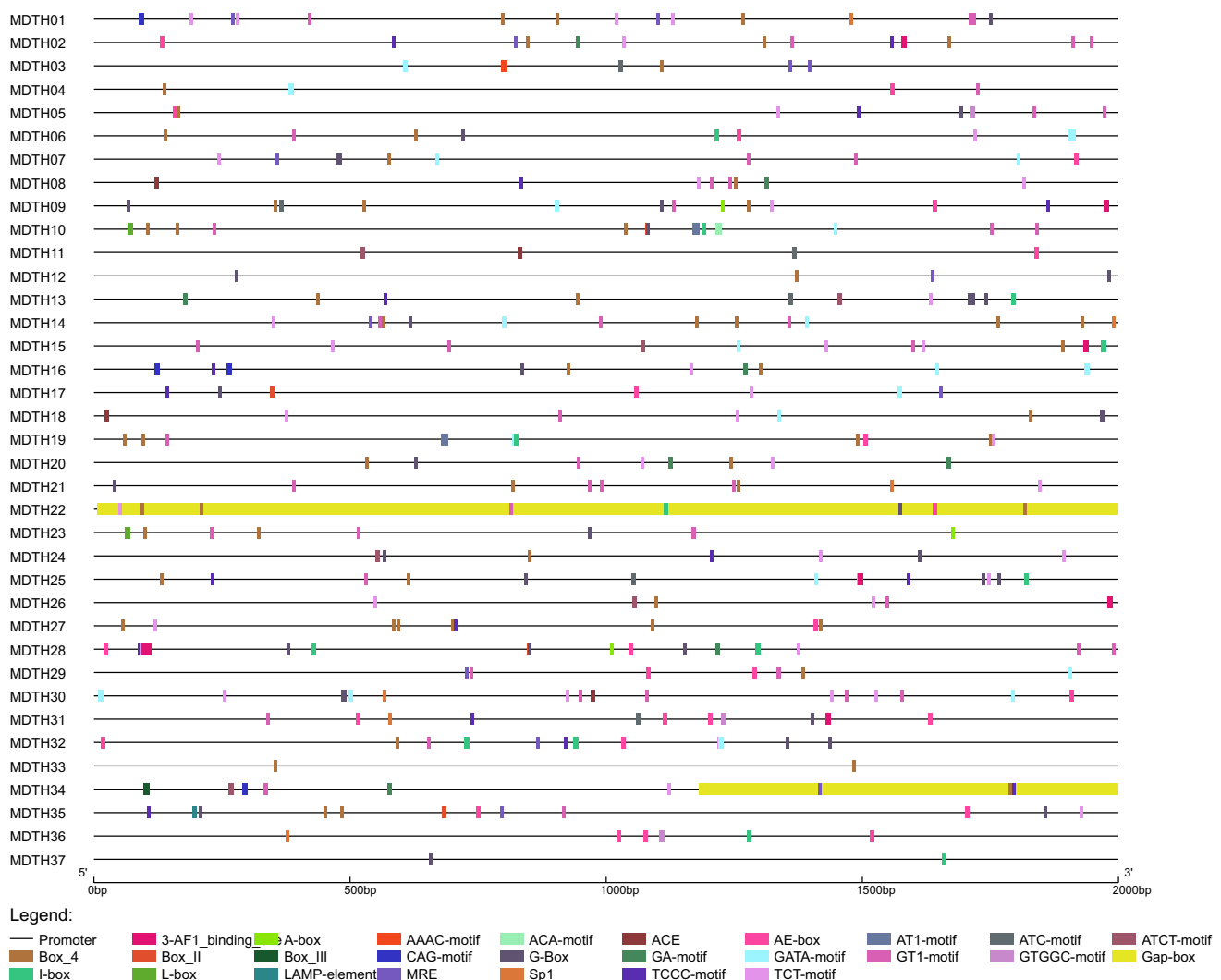


Fig. 5 Binding sites for the transcription factors involved in light-dependent signaling pathways found in the promoter regions of MdTH genes

WRE motifs found in the regulatory regions of 35.1%, 18.9%, 43.2%, and 5.4% genes, respectively. At least one of these elements contains the regulatory regions of 67.6% MdTH genes (Fig. 6).

The promoter regions of each MdTH contain binding sites for regulatory proteins with the MYB domain. They are involved in the regulation of biotic and abiotic stress responses, epigenetic control, hormonal signaling pathways, the regulation of cell differentiation and shape, and phenylpropanoid biosynthesis. It was found that 97.3% of MdTH promoters contain sequences for binding transcription factors of the MYC family. They are part of the hormonal regulation system for jasmonic and salicylic acids participating in coordination of plant growth and development in response to various stress types and other biological processes. 54.1% of regulatory regions contain W-box, a DNA sequence binding WRKY transcription factors. They

perform many functions, such as the formation of resistance to pathogens and stress; they are involved in hormonal regulation, ontogenesis, etc. (Fig. 6). Induction of the expression of trihelix transcription factors may be caused by plant hormones: abscisic acid, which is evidenced by the presence of at least one of the following elements: ABRE, ABRE2, ABRE3a, ABRE4, and AT-ABRE in the regulatory regions of 91.9% of genes; gibberellin (at least one of p-box, TACT-box, and GARE elements found in 75.7%), salicylic acid (TCA and/or *as-1* elements have the regulatory regions of 91.9% of genes), ethylene (ERE elements were found in 48.6% of cases). CGTCA and TGACG sequences involved in the methyl jasmonate-induced signaling pathway were found in the promoter regions of 75.7% and 83.8% of genes, respectively. The elements associated with hormone-regulated signaling cascades were found in the promoter regions of each MdTH (Fig. 6). MdTH expression may be

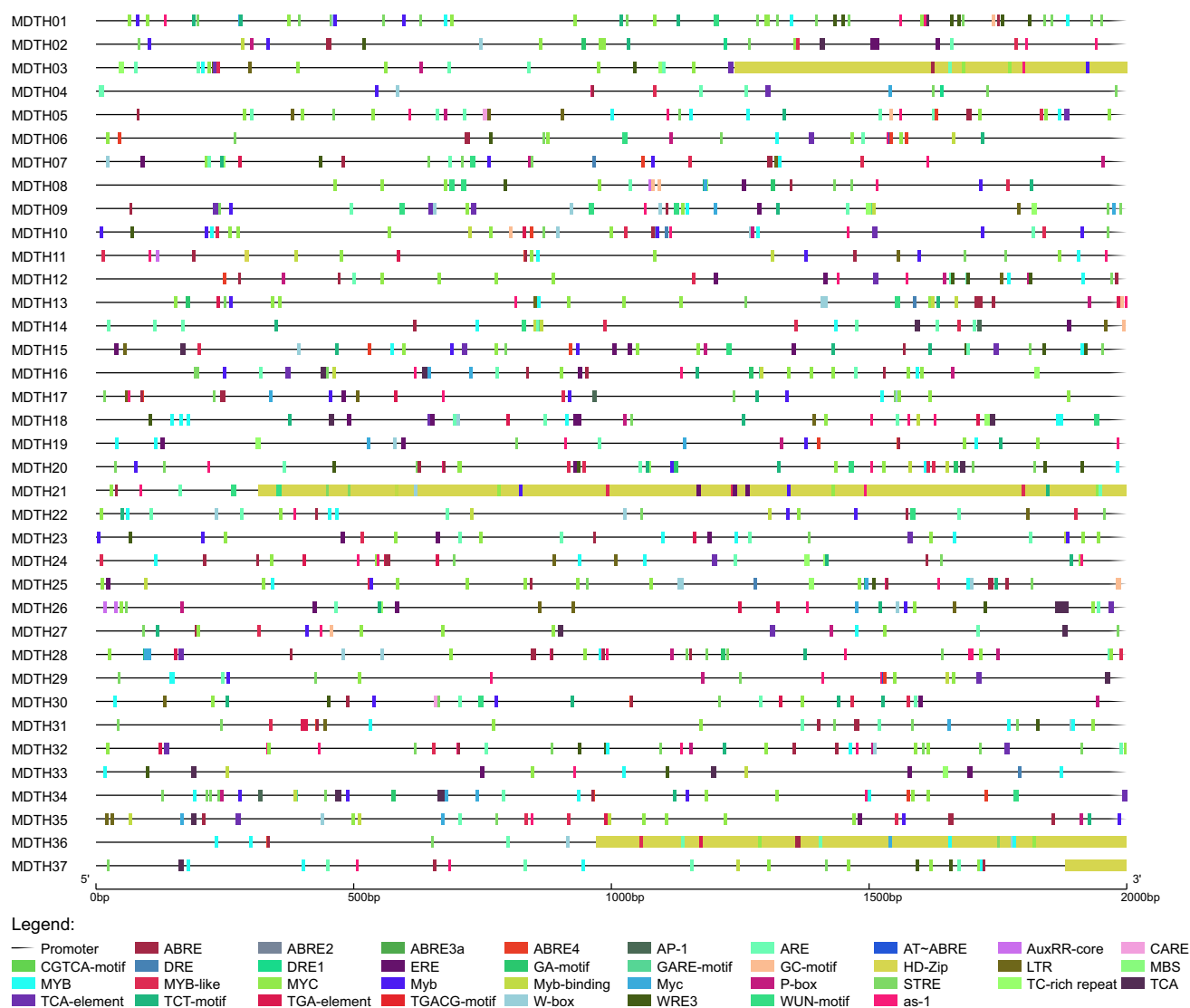


Fig. 6 Binding sites of the transcription factors involved in stress- and hormone-induced signaling pathways found in MdTH promoter regions

tissue-specific and may depend on the plant development stage. This is evidenced by AC1, AC2, telo-box, and F-box elements (the latter is also involved in biotic and abiotic stress responses); by the dOCT element, which is involved in the regulation of growth and development, motif I (expression in the root). The expression of trihelix transcription factors may be also regulated by the following elements: AP-1, which regulates flowering, O2—prolamin metabolism, and MBS-I—flavonoid biosynthesis; GCN-4, which is associated with the endosperm expression, and CAT—with meristem; HD-ZIP1, which is responsible for palisade mesophyll differentiation. Apparently, a list of MdTH functions also includes the regulation of proliferation and a cell cycle, as evidenced by the regulatory elements re2f-1, NON, MSA-like, and e2fb.

Thus, analysis of the nucleotide structure of regions preceding MdTH suggests that they are involved in a complex regulatory network, which ensures plant growth and development. Analysis of the detected regulatory elements suggests that MdTH are transcribed continuously. However, various conditions may lead to a change in their level. It may be affected by ontogeny stages, the specifics of tissues and organs, various environmental factors, including changes in light intensity, as well as various types of biotic and abiotic stress effects. The results obtained are consistent with those obtained when studying the expression of trihelix proteins in other plants, demonstrating that most of them are expressed under normal conditions, but their expression levels may change significantly under stress.

Phosphorylation sites specific for mitogen-activated protein kinase (MAPK) and miRNA targets prediction

The paper (Li et al. 2015) demonstrates that, in *Arabidopsis*, after treatment with MAMP (microbe-associated molecular patterns), the MAP KINASE4 enzyme triggered the rapid phosphorylation of the *ASR3* trihelix gene (Li et al. 2015). This suggests that the members of this family may be post-translationally regulated by phosphorylation. Analysis of hypothetical MdTH proteins showed that each of them has at least one putative phosphorylation site.

Moreover, serine residues predominate among them. Among GT-2 subfamily members, the number of putative phosphorylated serine residues ranges from 1 to 19 (more than 10 in most members) and from 0 to 5 in the case of threonine. Among SH4 subfamily members, from 2 to 49 serine residues and from 0 to 7 threonine residues can be phosphorylated. For SIP1 members, the number of putative serine phosphorylation sites ranges from 7 to 22, and threonine—from 0 to 4. For GT-1 subfamily members, the number of putative phosphorylated serine residues ranges from 6 to 11, and threonine—from 1 to 3. And in the case of GT γ family members, which have from 13 to 15 hypothetical serine phosphorylation sites, no threonine phosphorylation site was found. Representatives of GT δ have three putative phosphorylated threonine residues and from 31 to 40 serine residues each.

MicroRNAs are a class of non-coding regulatory RNAs, which are involved in the regulation of gene expression by inhibiting target mRNA translation or cleavage (Unver et al. 2009; Eldem et al. 2013; Zhang 2015). In apple, miRNAs play an important role in the development, stress, and pathogen attack responses, and may regulate the gene expression through target cleavage. For example, in response to an infection of apple with the Valsa canker causative agent, 353 differentially expressed microRNAs were found, which may target 1077 unigenes with 2251 cleavage sites (Feng et al. 2017) involved in a response to the infection of the pathogen *Valsa mali*. When elucidating the role of individual microRNAs, it was shown that *MdmiR285N* has binding sites in 17 different transcription factor families, including C2H2, CSD, HD-ZIP, NAC, and WRKY, which play an important role in plant development and stress response (Pompili et al. 2020).

It was found out that 21 trihelix genes have one or more targets for binding to miRNAs. In total, 75 types of miRNAs were identified. They are involved in the inhibition of mRNA translation or cleavage. The largest number of binding sites for various miRNAs (32) was found in the mRNA sequence MdTH34. MdTH12 and MdTH17 contain 19 binding sites each; MdTH36 and MdTH8—six each; MdTH5 and 15—five each; MdTH35 has four; MdTH7, MdTH29, and MdTH30 have three each. MdTH4, MdTH6, and MdTH31 contain two binding sites each. MdTH1, MdTH9, MdTH14,

MdTH20, MdTH28, MdTH32, and MdTH37 have one. At that, MdTH34 contains two copies of binding sites, *mdm-miR535b* and *mdm-miR535c*, while all other family members contain only one copy of each binding site.

Expression profiles of the MdTH gene in response to abiotic stresses

Various adverse environmental factors, such as drought, high and low temperatures, soil salinity, may affect the growth, development, and productivity of plants. To get adapted to stress conditions, plants may change the expression of stress-related genes. In this work, we have tested response of 14 MdTH genes to some unfavorable abiotic factors, since trihelix family members are known to be involved in adaptation processes in other plants (Fang et al. 2010; Li et al. 2019; Murata et al. 2002; Xie et al. 2009).

Evaluation of the expression of 14 MdTH genes in response to drought (Fig. 7) showed that 10 out of them are upregulated, most significantly—MdTH4 at the 24th hour of exposure. However, MdTH21, MdTH23, and MdTH30 were downregulated when exposed to drought.

Low temperature (4 °C) caused a significant decrease in the MdTH4 relative expression level after 2 h of exposure and MdTH30 after 4 h (Fig. 7). Most genes (MdTH8, MdTH9, MdTH20, MdTH21, MdTH31, MdTH35, and MdTH36) were upregulated after 2 and/or 4 h of exposure.

During the experiment, it was found that after 24 h of exposure, most of the genes exhibited downregulation, with the relative expression level below the baseline at the time of sampling. However, there were exceptions to this trend. For instance, the relative expression level of the MdTH35 gene was higher after 24 h of exposure than at the time of initial sampling.

To assess the effect of high temperature, the plants were placed in a climate chamber at 40 °C. Evaluation of the expression profiles of the MdTH genes in MM-106 rootstocks (Fig. 7) showed that the transcription level of most investigated genes had changed insignificantly, and some of them (MdTH4, MdTH11, MdTH22, etc.) were downregulated. MdTH8, MdTH20, and MdTH36 were upregulated after 24 and 4 h of exposure; the most significant change occurred in the MdTH36 relative expression in response to high temperature.

Out of all the stress factors examined, exposure to saline solution produced the most similar response among the genes under study. After 2-h exposure, most genes (MdTH4, MdTH8, MdTH9, MdTH11, MdTH20, MdTH21, MdTH22, MdTH24, MdTH30, MdTH31, and MdTH36) were upregulated, although their response rates varied. Specifically, MdTH4 and MdTH24 exhibited higher levels of transcription after exposure to saline solution.

Fig. 7 MdTH gene expression profiles under the conditions of drought (A), low temperatures (B), high temperatures (C), and salinity (D) of the MM-106 apple variety with measurement points at the 0, 2nd, 4th, and 24th hour. The data were normalized in relation to the apple housekeeping gene *Efl-α*. The vertical stripes show the standard deviation

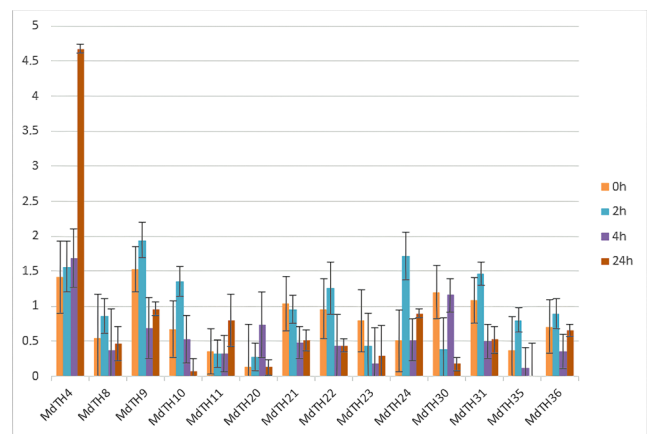
Discussion

In this work, 37 apple trihelix genes were identified using the HMM-based search. The presence of a specific Myb/SANT-LIKE domain was confirmed by the SMART algorithm. The tomato trihelix family also contains 37 genes, whereas other plants may have more or fewer (Yu et al. 2015). For example, 94 trihelix members were identified in the wheat genome, 71—in the soybean, 52—in *Brassica rapa*, 41—in rice, 31—in *Tatar buckwheat*, and 20—in chrysanthemum (Xiao et al. 2019; Liu et al. 2020; Wang et al. 2017; Li et al. 2019; Ma et al. 2019; Song et al. 2016). The number of genes in the trihelix family may depend on various polyploidization processes, which some species went through during evolution.

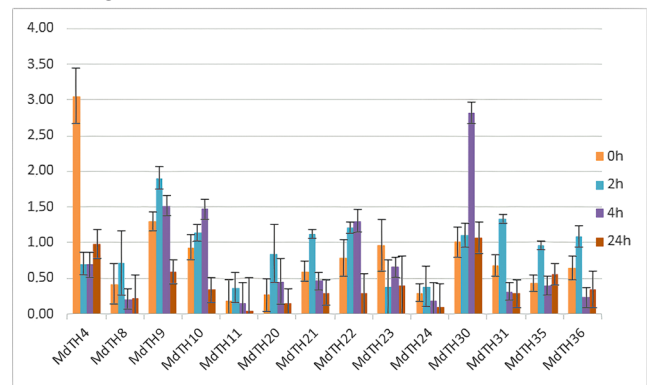
The hexaploid wheat genome was formed through the interspecies hybridization of three diploid species. As a result, trihelix genes now have homologous copies in the A, B, and D genomes of *T. aestivum*. In contrast, diploid *T. uratru* has only 22 members of this gene family (Xiao et al. 2019).

Duplication of the ancestral genome plays an important role in the formation of the domestic apple genome (Velasco et al. 2010). In this regard, the gene pairs located on the chromosomes of common origin are characterized by high sequence identity (Fig. 2), as well as exon-intron structure and the distribution of conserved amino acid motifs (Fig. 4).

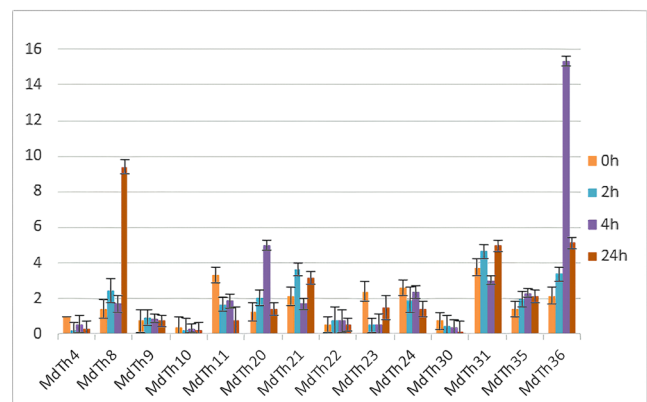
MdTH are members of six subfamilies (GT-1, GT-2, SH4, GTδ, GTγ, and SIP1). Members of five subfamilies are always identified among the trihelix genes of different species, while GTδ members are not present in all plants—they have been found in sunflower, tomato, and rice (Song et al. 2021; Yu et al. 2015; Li et al. 2019). GTδ subfamily members have differences in their exon-intron structure and in the distribution of conserved amino acid motifs compared with the members of other subfamilies. They have more introns, and typical conserved motifs are located at the C-terminus of the amino acid sequences (Fig. 4) (Song et al. 2021). In general, members of the same subfamily might share similar gene structure (Fig. 4). The locations of conserved amino acid motifs are similar to phylogenetic analysis results. It is typical of the trihelix transcription factors identified in other species (Xiao et al. 2019; Liu et al. 2020; Wang et al. 2017; Li et al. 2019; Ma et al. 2019; Song et al. 2016; Song et al. 2021). Genes, which are similar in structure, may perform similar functions.



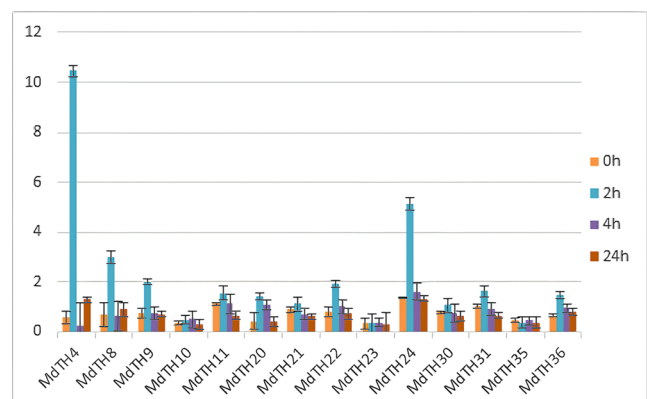
A Drought



B Cold



C Heat



D Salinization

Trihelix genes differ in their exon-intron structure. In the study presented here, both intronless genes and the genes with the number of introns varying from one to sixteen were found in apple. GT γ members are intronless not only in apple, but also in wheat, pineapple, *Brachypodium distachyon*, and tomato (Xiao et al. 2019; Wang et al. 2022; Wang et al. 2019; Yu et al. 2015). The maximum number of introns was found in the members of the GT δ subfamily of apple, sunflower, tomato, and rice (Yu et al. 2015; Li et al. 2019; Song et al. 2021). The exon-intron organization similarity of the same subfamily members was also noted for the trihelix transcription factors of other plants, e.g., buckwheat *Fagopyrum tataricum* (Ma et al. 2019). Similar to the apple, the number of introns in buckwheat trihelix genes ranged from one to sixteen, and the genes with the fewest number of introns were the members of GT γ and SIP1.

Analysis of the promoter regions of MdTH genes showed that their expression can be regulated by light. The regulatory region of each gene contains from 2 to 10 types of elements involved in light-induced signaling cascades. Induction of the expression of MdTH genes may be triggered by various phytohormones such as abscisic acid, gibberellin, salicylic acid, ethylene, and methyl jasmonate. Elements associated with hormone-regulated signaling cascades were found in the promoter regions of each gene. Plant hormones participate in regulating many processes. For example, abscisic acid contributes to regulating plant growth and development, stress responses, leaf aging, and bud germination, seed germination, and stomatal closure, among others, through transcriptional and post-transcriptional mechanisms (Chen et al. 2020). Gibberellic acid stimulates various trigger transitions in plant ontogenesis, such as the transition from seed dormancy to germination, from meristems to shoot growth; the transition of leaves from juvenile to mature stages, from their vegetative stage to flowering, and it also participates in signaling cascades initiated by changes in lighting, temperature, and water regimes (Gupta and Chakrabarty 2013). Salicylic acid activates protective reactions in plants and regulates their growth; it also contributes to regulating their immunity, resistance to biotic and abiotic stresses, and interaction with soil microbiome (Koo et al. 2020). Methyl jasmonate, jasmonic acid, and its amino acid conjugates, collectively known as jasmonates, are important cellular regulators, which mediate various developmental processes, including root growth, pollen production, and plant resistance to insects and pathogens (Creelman and Mullet 1997; Kessler and Baldwin 2002). For example, when *Nicotiana attenuata* is eaten by herbivores, methyl jasmonates contribute to regulating the synthesis of volatile substances, which attract natural enemies of herbivores (Kessler and Baldwin 2001; Mattiacci et al. 1995), and secondary metabolites acting as direct means of protection, such as neurotoxin nicotine (Baldwin 1999). Ethylene influences various plant developmental processes, e.g., seed germination, fruit ripening, aging, and abscission, as well

as responses to various types of stress such as flooding, salinization, and soil compaction (Iqbal et al. 2017; Binder 2020). The presence of regulatory elements for hormonal regulation in MdTH promoter regions may indicate that they are possibly involved in such processes.

In the promoter region of each MdTH gene, there was at least one element, indicating their expression under stress conditions, such as anaerobic stress, drought, low temperature, pathogen attack, and injury. Therefore, it is expected that the expression of MdTH genes will vary across different plant organs and may change depending on the stage of ontogenesis, as well as external environmental factors, including light intensity, temperature regime, oxygen and salt levels in soil, and pathogen or herbivore attacks.

As immobile organisms, plants cannot avoid the impacts of abiotic stress. Genes, contributing to their increased resistance, play an important role in the process of plant adaptation to stress. Previous studies showed that trihelix genes respond to such stress factors as over-hydration, salinity, and other adverse effects, as well as hormone treatments involved in a stress response like abscisic acid (Fang et al. 2010; Li et al. 2019; Murata et al. 2002; Xie et al. 2009; Liu et al. 2020). In our work, we demonstrated that drought, salinity, high and low temperatures affect the expression of MdTH genes. Among the genes studied, MdTH4 and MdTH24, the members of SH4 and SIP1 subfamilies respectively, showed the greatest increase in their expression level under soil salinity. The MdTH4 gene also exhibited the highest relative increase in the expression level during drought. During a decrease in temperature, the biggest change in the expression level was observed in MdTH30 of the GT-2 cluster. Interestingly, the same gene was among the three the expression levels of which significantly decreased under drought stress. When the temperature rose, the expression level of MdTH8, MdTH20, and MdTH36 showed the highest increase. These genes are the members of the GT γ , GT-2, and GT-1 subfamilies respectively.

Representatives of various MdTH subfamilies have shown the ability to respond to stress conditions. Notably, an increase in relative expression levels of MdTH4, MdTH24, MdTH30, MdTH8, MdTH20, and MdTH36 under stress conditions suggests their potential involvement in responding to the corresponding types of abiotic stress.

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Data Availability The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

Declarations

Competing interests The authors declare no competing interests.

Human and animal rights and informed consent This article does not contain any studies with human participants or animals performed by any of the authors.

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